

Outset · AI in Healthcare · Day 3 · Group 6

Does estrogen keep the mind sharp?

A twenty-year lesson in confounding, and why a survey can hint but only an experiment can prove

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BACKGROUND

A twenty-year detective story

THE EXCITING CLUE (1996-2002)

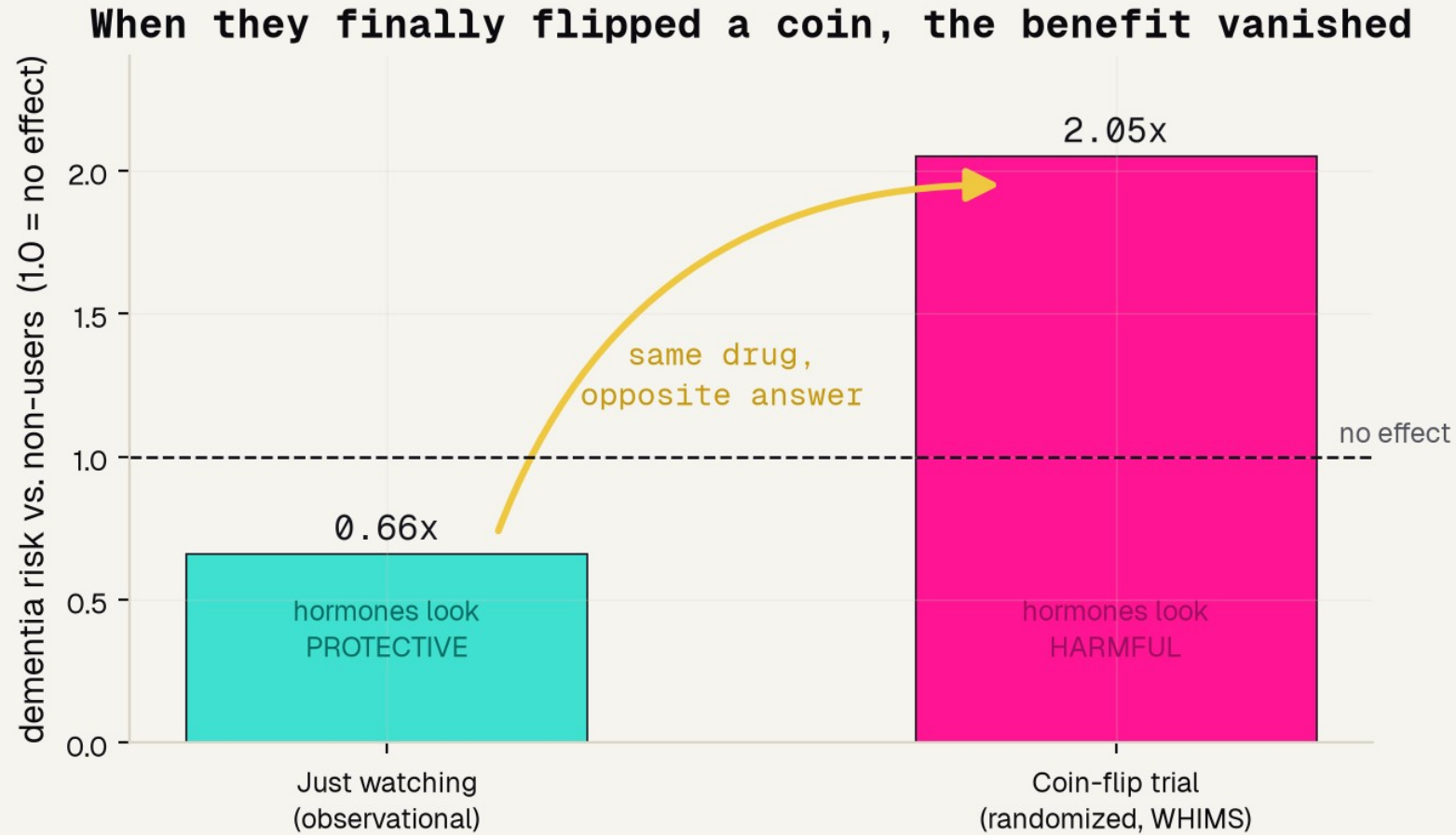
Studies that simply WATCHED women found that estrogen users developed Alzheimer's less often and later [1, 2]. It looked like solid proof the pills protected the brain.

THE PLOT TWIST (2003)

Then the WHIMS trial let a COIN FLIP decide who got hormones. The benefit vanished, and dementia risk actually ROSE in women 65+ [3, 4].

Same drug, opposite answer. The rest of this talk is one question: HOW?

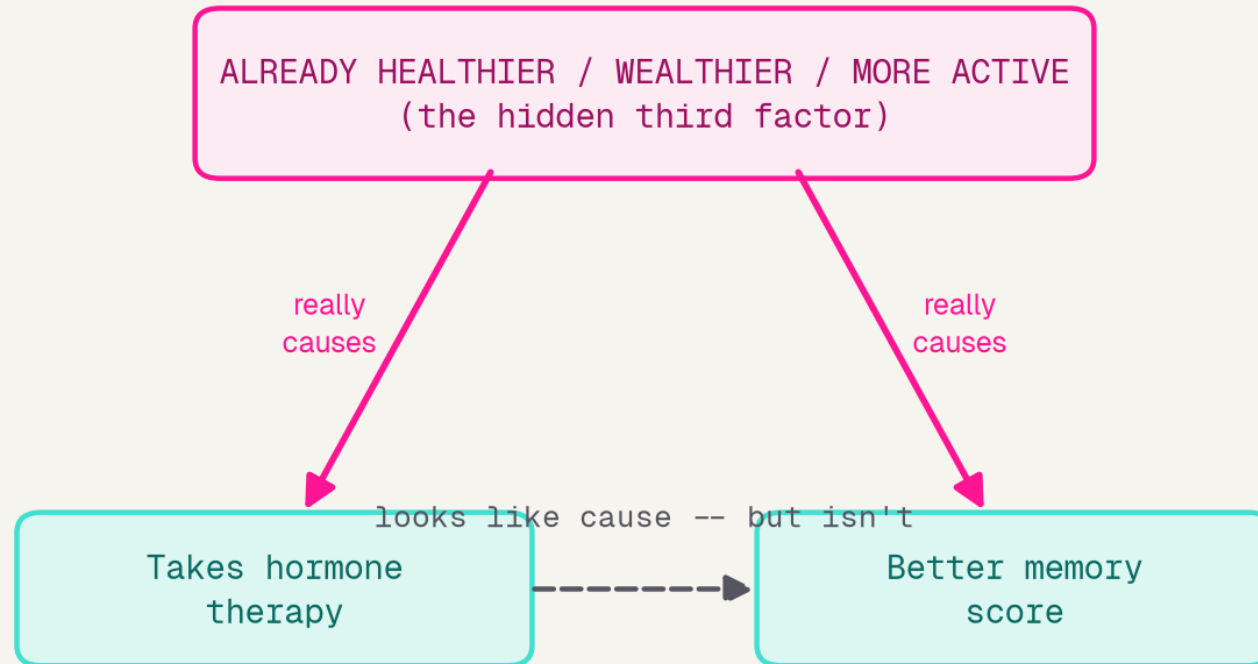
When they flipped a coin, the benefit vanished



adapted from Tang 1996 / Zandi 2002 (observational) vs. Shumaker 2003 WHIMS (randomized)
Just watching (turquoise) said hormones protect; the coin-flip trial (pink) said they harm.

Healthy-user bias: a hidden third factor

Healthy-user bias: a hidden factor fakes the link



concept -- adapted from Wharton 2009 / Vandenbroucke 2009

Who chose the pill?

Users were richer, more active, and saw doctors more often to begin with [5].

The hidden factor

That same head start drives BOTH the drug use AND the good memory.

So the bottom link

only LOOKS like cause. It is a mirage -- confounding, a.k.a. healthy-user bias [6].

Ice cream and shark attacks

On hot days, ice-cream sales and shark attacks BOTH rise together. Does ice cream summon sharks?

THE ANALOGY

No -- the hot SUMMER WEATHER drives both: heat sells ice cream and sends people into the ocean. Neither one causes the other.

OUR VERSION

Estrogen use and good memory rise together. Being HEALTHIER to start with drives both -- the pill may summon nothing.

When two things move together, hunt for the hidden third factor before you believe one caused the other.

METHODS

Our data: NHANES women 60 and older

739 WOMEN

all age 60+, drawn from the NHANES 2013-14 US national health survey.

USED ESTROGEN?

whether she ever took female hormones -- yes or no, self-reported.

HER BACKGROUND

age, education, and family income -- the confounders we must check.

MEMORY SCORE

a digit-symbol test; higher = sharper. We predict the weaker-scoring half.

The one question we can actually ask

We cannot randomly assign estrogen in a survey. So we ask a narrower, honest question:

Does the estrogen effect SURVIVE once we account for age, education, and income?

CRUDE

Estrogen's effect all by itself -- ignoring everything else about the woman.

ADJUSTED

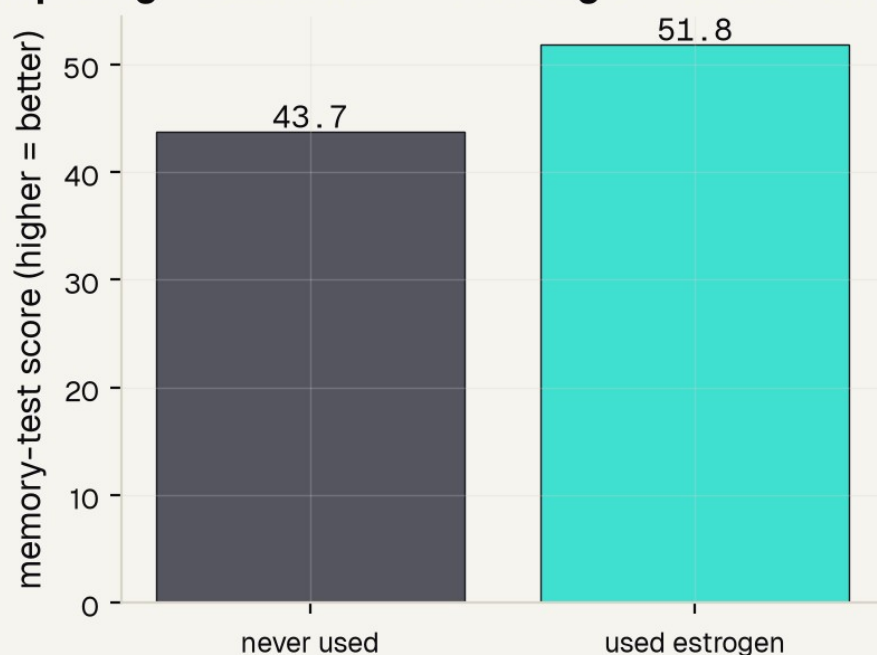
Estrogen's effect AFTER accounting for age, education, and income.

If the effect shrinks from crude to adjusted, the headline was mostly confounding -- not cause.

RESULTS

The tempting headline

The tempting headline: 'estrogen users think better'

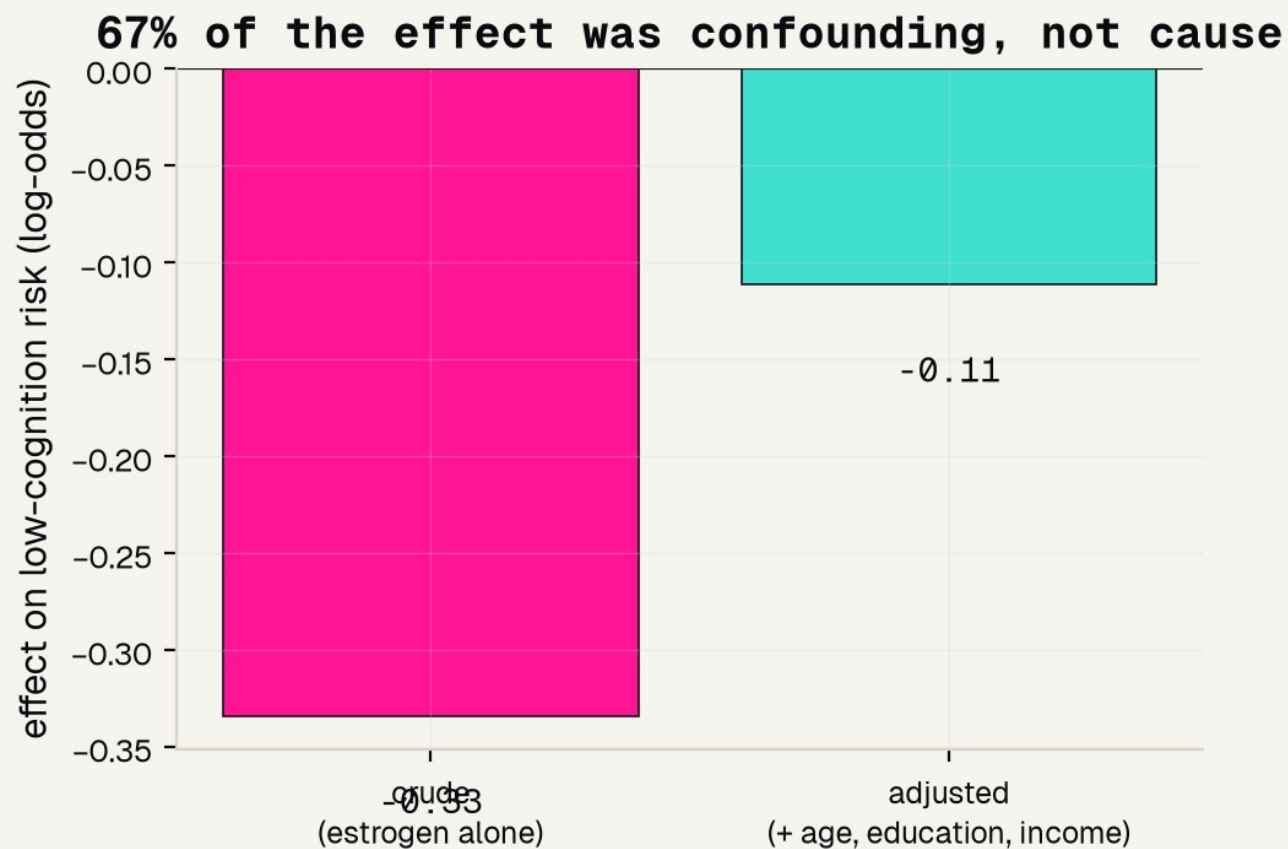


+8.1

points -- the raw gap, users over non-users

51.8 versus 43.7. Taken at face value, the exciting headline writes itself: 'estrogen users think better.' But we never checked whether the two groups were the same kind of people to begin with.

Two-thirds of the effect was confounding



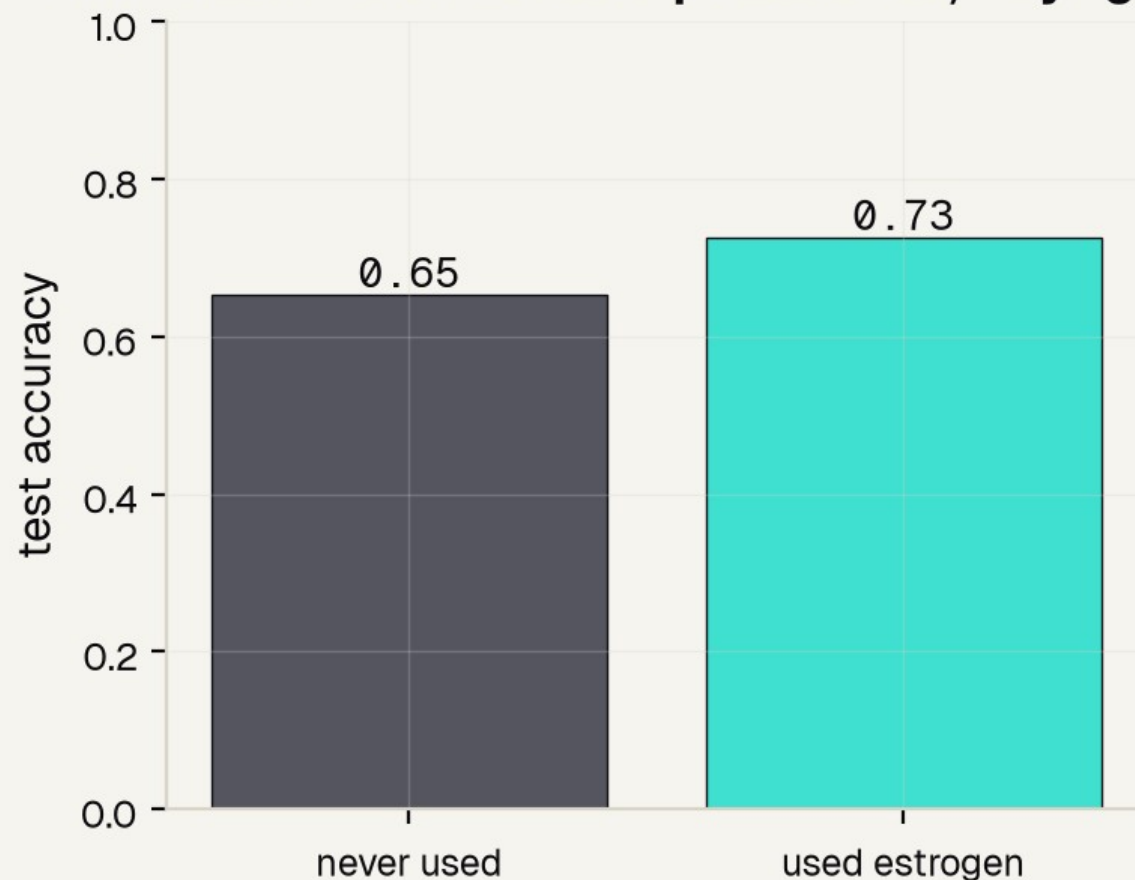
67%

of the effect was confounding

The crude effect (-0.33) shrank to just -0.11 once we adjusted for age, education, and income. That missing two-thirds was never the estrogen -- it was that users were healthier and better-off to start with.

Does the model work for everyone?

How well the model predicts, by group



0.73

test accuracy for estrogen-USERS

0.65

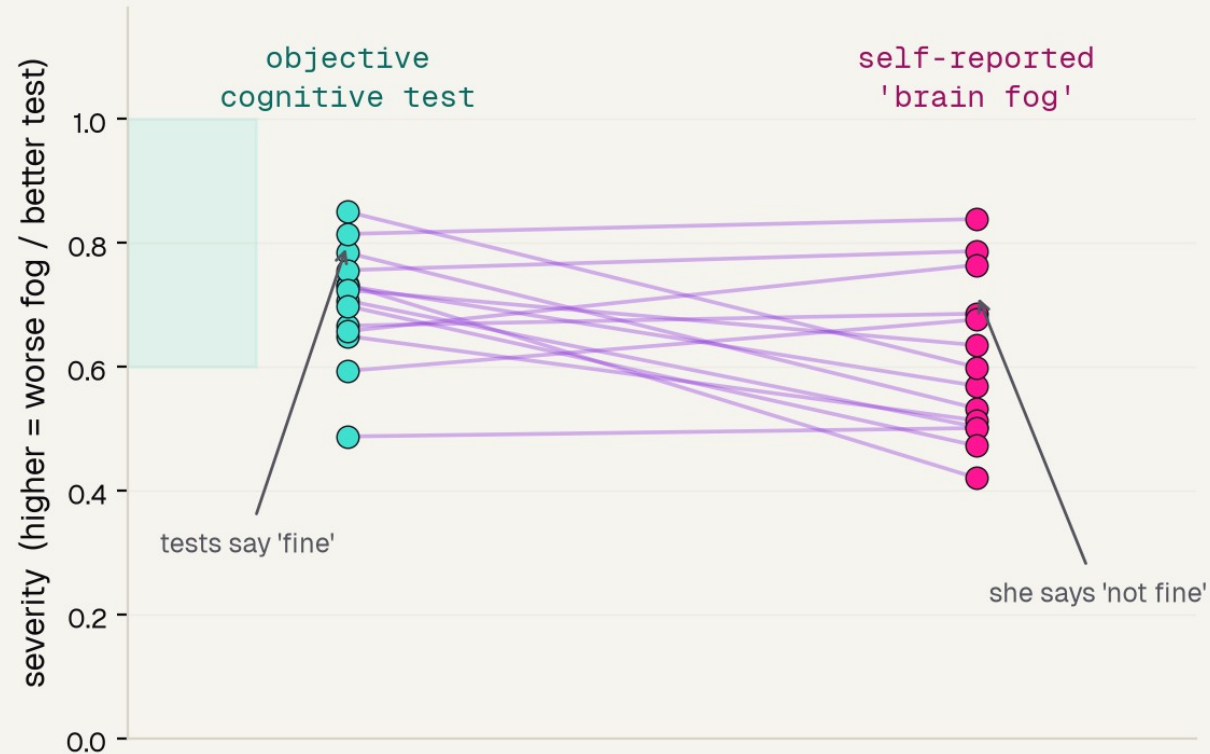
test accuracy for NON-users

A tool that is reliably right for one group and shakier for another can quietly do harm if a clinic trusts it equally. Naming the gap out loud is the honest move.

WHAT THE NUMBERS CAN'T CAPTURE

The tests say “fine”; she says “not fine”

The tests can say 'normal' while the patient feels foggy



adapted from Maki 2024, Harvard Health (menopause & brain fog)

The mismatch

Many women report real, daily brain fog even when the objective test looks perfectly normal [7].

Why it matters

Women's symptoms are too often waved off as stress or emotion instead of investigated [8].

Hold both ideas

The confounding is real AND the patient's account is real evidence -- not noise to dismiss.

A survey can hint; only an experiment proves

WHAT IT CAN DO

Show a tempting headline is mostly a mirage, and that adjusting shrinks the effect by ~67%.

WHAT IT CANNOT DO

Prove cause. We can only correct for confounders we actually measured and thought of.

WHAT SETTLES IT

A randomized trial: a coin flip, not the patient, assigns the drug. That is why WHIMS could overturn 20 years of headlines.

References

[1] Tang et al. 1996, Lancet -- observational; estrogen users got Alzheimer's less/later.

[2] Zandi et al. 2002, JAMA (Cache County) -- observational; long-term users lower risk.

[3] Shumaker et al. 2003, JAMA -- WHIMS RCT; hormones DOUBLED dementia risk in 65+.

[4] Rapp et al. 2003, JAMA -- WHIMS RCT; no cognitive benefit, scores trended worse.

[5] Wharton et al. 2009, Maturitas -- healthy-user bias; users start out healthier.

[6] Vandenbroucke 2009, Lancet -- why observational studies and RCTs disagreed.

[7] Maki 2024, Harvard Health -- menopause 'brain fog' vs. normal test scores.

[8] Chen & Shafir 2025, Harvard Health -- the dismissal of women's symptoms.

The one habit to carry out the door

Suspect the chooser, not the treatment.

When a group that CHOSE a treatment looks healthier, the head start usually belongs to the person, not the pill -- and only a coin-flip experiment can promote a correlation to a cause.

And take the patient's account seriously -- even when the test says 'fine.'

Thank You

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